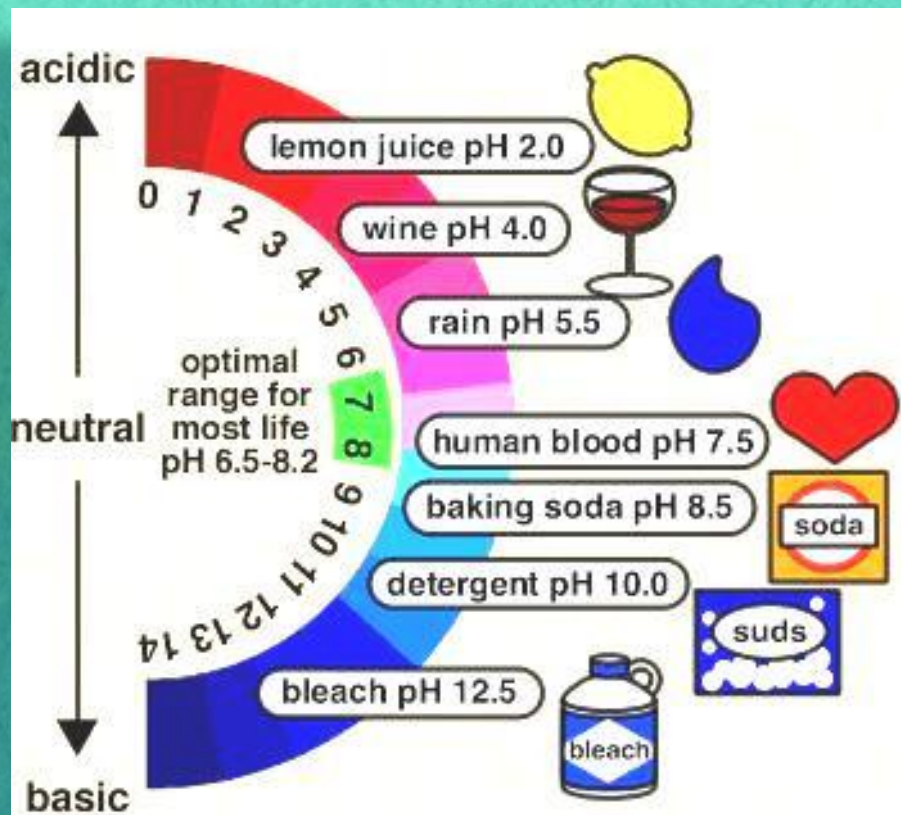


Reactions Involving Acids and Bases



Arrhenius Definition

Acids: any substance that produces hydrogen ions in solution

Bases: any substance that produces hydroxide ions in solution

As part of your homework:

Review acids and bases in the text and be able to define strong vs weak acids and bases

Producing Acids & Bases

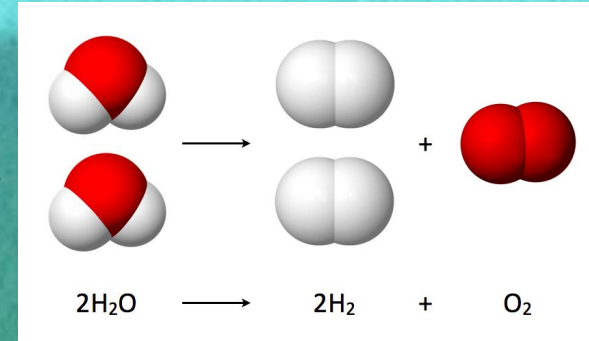


- need $\text{H}^+_{(\text{aq})}$ or $\text{OH}^-_{(\text{aq})}$ to be readily available
- and capable of interacting with other chemicals
- Their reactivity often makes them difficult to store so we need methods of producing these



Producing Acids & Bases

- need something strong enough to break H_2O or HOH into $\text{H}^+_{(\text{aq})}$ or $\text{OH}^-_{(\text{aq})}$
- but it can't be too reactive to ensure it persists
- and it can only allow either $\text{H}^+_{(\text{aq})}$ or $\text{OH}^-_{(\text{aq})}$ to continue to exist or the result is still neutral
- the answer = **OXIDES**
- incredibly stable but contains reactive oxygen, O^{2-}





Non-metal Oxides

- are covalent molecules such as $\text{SO}_{2(g)}$ or $\text{CO}_{2(g)}$
- react directly with water as a synthesis reaction
- e.g. $\text{SO}_{2(g)} + \text{H}_2\text{O}_{(l)} \rightarrow \text{H}_2\text{SO}_{3(aq)}$ or $\text{H}_2\text{SO}_{2(aq)}$, etc.
- non-metal oxides + water $\rightarrow$ **ACIDS**



Metal Oxides

- mostly ionic, e.g. $\text{MgO}_{(g)}$ which dissolve in $\text{H}_2\text{O}_{(l)}$
- so overall: $\text{MgO}_{(g)} + \text{H}_2\text{O}_{(l)} \rightarrow \text{Mg}^{2+}_{(aq)} + 2 \text{OH}^{-}_{(aq)}$
- metal oxides + water $\rightarrow$ **BASES**

Neutralization

Acid: H^+ + anion

Base: cation + OH^-

Acid + Base = HOH + salt

cation + anion

- is a double displace reaction
- in which the acidic and basic properties cancel each other
- e.g. $\text{HCl}_{(\text{aq})} + \text{NaOH}_{(\text{aq})} \rightarrow \text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$
- recall, the net ionic equation clearly indicated the neutralization reaction
- $\text{H}^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})} \rightarrow \text{H}_2\text{O}_{(\text{l})}$

Carbonate Neutralization

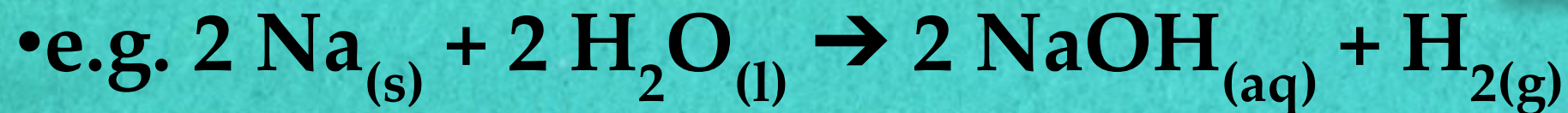
- carbonates react with acids producing carbonic acid and a salt
- e.g. $2 \text{HNO}_{3(\text{aq})} + \text{Na}_2\text{CO}_{3(\text{aq})} \rightarrow \text{H}_2\text{CO}_{3(\text{aq})} + \text{NaCl}_{(\text{aq})}$
- but carbonic acid is not stable in water so it immediately decomposes
- so $\text{H}_2\text{CO}_{3(\text{aq})} \rightarrow \text{CO}_{2(\text{g})} + \text{H}_2\text{O}_{(\text{l})}$
- $2 \text{HNO}_{3(\text{aq})} + \text{Na}_2\text{CO}_{3(\text{aq})} \rightarrow \text{CO}_{2(\text{g})} + \text{H}_2\text{O}_{(\text{l})} + \text{NaCl}_{(\text{aq})}$



THE END

Reactive Metals

- a single displacement reaction
- a metal more reactive than hydrogen
- displaces one H in $\text{H}_2\text{O}_{(l)}$
- to directly create a **BASE** and hydrogen gas



potassium	most reactive	K
sodium		Na
calcium		Ca
magnesium		Mg
aluminium		Al
carbon		C
zinc		Zn
iron		Fe
tin		Sn
lead		Pb
hydrogen		H
copper		Cu
silver		Ag
gold		Au

Double Displacement

- a double displace reaction
- can be used to generate stronger acids
- as long as unwanted ions are locked in
- e.g. $2 \text{HI}_{(\text{aq})} + \text{Pb}(\text{NO}_3)_2(\text{aq}) \rightarrow 2 \text{HNO}_{3(\text{aq})} + \text{PbI}_{2(\text{s})}$

